MMF2030H — Machine Learning

Assignment 1

DUE DATE	October 27, 2026, 11:59 PM (before Lecture 4)
WEIGHT	20% of final grade
COVERS	Lectures 1–2 — risk minimization, gradient descent, bias–variance, linear models & KNN, regularization as MAP estimation, purged/embargoed CV, evaluation & calibration
FORMAT	Theory & derivation only — no coding component. Assignment 2 is where applied/coding work is assessed.
SUBMISSION	Single PDF via course portal; individual work

Instructions

- This assignment is completed individually. You may discuss concepts with classmates, but all derivations and write-up must be your own.
- Answer each question concisely but show your full work — a page or two of derivation per question is normal. Partial credit is given for a correct setup even if the final algebra has an error.
- Submit a single PDF with your three answers in order, clearly labeled A1–A3. State any assumptions you make.

Part A — Theory & Derivation (20 marks)

A1. Bias–variance and regularization as MAP estimation (7 marks).

(a) State the bias–variance decomposition of expected test-set squared error, $E[(y - \hat{f}(x))^2] = \mathrm{Var}(\hat{f}(x)) + \mathrm{Bias}(\hat{f}(x))^2 + \sigma^2$ (you do not need to re-derive it — the derivation and visual are in the Lecture 1 slides). Use it to explain, in one or two sentences each, why

- **i.** 1-nearest-neighbor regression has high variance and low bias
- **ii.** a linear regression fit on a small, noisy sample can have low variance but high bias if the true relationship is nonlinear.

(b) You are given a Gaussian likelihood $y \mid \beta \sim N(X\beta, \sigma^2 I)$ and an independent Laplace prior on each coefficient, $\beta_j \sim \mathrm{Laplace}(0, b)$, with density $p(\beta_j) = (1/2b)\exp(-|\beta_j|/b)$, so $\log p(\beta) = -\|\beta\|_1/b + \text{const}$ and the log-posterior (up to a constant) is $-(1/2\sigma^2)\|y - X\beta\|^2 - \|\beta\|_1/b$. Show that maximizing this is equivalent to the lasso estimator $\hat{\beta}_{\text{lasso}} = \arg\min_\beta \|y - X\beta\|^2 + \lambda\|\beta\|_1$, and give the explicit relationship between λ, σ^2, and b. (In lecture, we worked through the analogous derivation for ridge under a Gaussian prior — the same steps apply here, just with the Laplace log-density in place of the Gaussian.)

(c) Does the lasso penalty increase or decrease the variance of $\hat{\beta}$ across resamples, and does it increase or decrease bias? State both directions, then justify in one sentence by connecting to part (a).

A2. Evaluation methodology: cross-validation & calibration (9 marks).

(a) [5 marks] Purged & embargoed cross-validation. Consider 500 consecutive trading days (indexed 1–500); each day t's label is built with the triple-barrier method using a maximum horizon of 10 days, so day t's label can depend on prices through day t+10. A test fold spans days 200–250 inclusive, and the embargo length is set equal to the label horizon. State exactly which training days must be purged (their label windows overlap the test fold) and which further days must be embargoed. Then, in one or two sentences, explain why embargoing is needed in addition to purging, and contrast this fix with the walk-forward (expanding-window) alternative from the same lecture.

(b) [4 marks] Evaluation as a loss choice: calibration vs. discrimination. Define the Brier score and its decomposition BS = REL − RES + UNC (reliability/calibration, resolution, and uncertainty). You've built a model on 10 loan applicants, grouped by predicted default probability: 5 applicants were given predicted probability 0.40, of whom 0 actually defaulted; 5 applicants were given predicted probability 0.60, of whom all 5 actually defaulted. Compute REL, RES, UNC, and the overall Brier score for this model, and state what its discrimination (ranking ability) is. Then, in two to three sentences, explain concretely why, in a lending/credit-decision context, this model's calibration failure costs money even though its ranking is perfect.

A3. Gradient descent, derived and diagnosed (4 marks). Derive the gradient descent update rule $\theta_{t+1} = \theta_t - \alpha\nabla L(\theta_t)$ from a first-order Taylor expansion of L around θ_t. You are given the following fact: if ∇L is L-Lipschitz continuous ($\|\nabla L(\theta) - \nabla L(\theta')\| \le L\|\theta - \theta'\|$ for all θ, θ' — i.e., the gradient cannot change too abruptly), then the descent lemma guarantees L decreases at every step whenever the step size satisfies $0 < \alpha \le 1/L$. Using this fact together with the elongated-bowl loss surface from the Lecture 1 gradient descent visualization as your reference: (i) state the failure mode when α is too large, then justify it in one sentence using the descent-lemma condition; (ii) state the failure mode when α is too small, then justify it in one sentence; (iii) state whether feature scaling increases or decreases the surface's condition number, then justify in one sentence how that changes the usable range of α.

Deliverables Checklist

- Single PDF with your answers to A1–A3, clearly labeled and in order.

Assignment 1 - Jaime Coito Galdó

A1 a) The bias-variance decomposition is:

$$\text{Var}(\hat{f}(x)) + \text{Bias}(\hat{f}(x))^2 + \sigma^2$$

where variance measures how much the fitted model changes across different training samples, bias measures the systematic error between the model's average prediction and the true relationship, and σ^2 is irreducible noise.

i) 1-nearest-neighbor regression has low bias because it is extremely flexible and makes almost no assumptions about the functional form of the relationship. However, it has high variance because its prediction depend on a single training observation, so a small change in the training sample can substantially change the nearest neighbor and therefore the prediction.

ii) Linear regression can have high bias when the true relationship is nonlinear because it restricts the fitted function to a linear form and therefore cannot represent the true relationship correctly. Its restricted model class makes it less sensitive to changes in the training sample than a highly flexible method, giving it relatively low variance.

b) The MAP estimator maximizes the log-posterior:

$$\hat{\beta}_{MAP} = \arg\max_{\beta} \left[-\frac{1}{2\sigma^2}\|y - X\beta\|_2^2 - \frac{1}{b}\|\beta\|_1 \right]$$

Maximizing this negative expression is equivalent to minimizing:

$$\hat{\beta}_{MAP} = \arg\min_{\beta} \left[\frac{1}{2\sigma^2} \|y - X\beta\|_2^2 + \frac{1}{L} \|\beta\|_1 \right]$$

Multiplying the entire objective by the positive constant $2\sigma^2$ does not change its minimizer, giving:

$$\hat{\beta}_{MAP} = \arg\min_{\beta} \left[\|y - X\beta\|_2^2 + \frac{2\sigma^2}{L} \|\beta\|_1 \right]$$

Comparing with the Lasso objective:

$$\hat{\beta}_{Lasso} = \arg\min_{\beta} \left[\|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right]$$

we obtain:

$$\lambda = \frac{2\sigma^2}{L}$$

Therefore, MAP estimation under a Gaussian likelihood and independent Laplace priors on the coefficient is equivalent to Lasso regression.

c) The lasso penalty generally decreases variance and increases bias. By shrinking the estimated coefficients toward zero, and setting some coefficients exactly to zero, lasso restricts the flexibility of the fitted model. That makes the estimates less sensitive to changes in the training sample, reducing variance, but the systematic shrinkage away from the unregularized estimates introduces additional bias.

In terms of bias-variance decomposition, lasso deliberately accepts higher bias in exchange for lower variance, with the goal of reducing total expected error.

A2 | a) Each observation on day (t) has a label window extending through day (t+10). The test fold is days 200-250.

A training observation before the test fold must be purged whenever:

$$t + 10 \geq 200 \Leftrightarrow t \geq 190$$

So the training observations on days 190-199 must be purged. Day 189 is still safe because its label window ends on day 199. The test fold ends on day 250, and the embargo length is 10 days. Therefore, the additional embargoed observations are 251-260. Thus, the split is

→ Train: 1-189
→ Purge: 190-199
→ Test: 200-250
→ Embargo: 251-260
→ Train: 261-500

Purging removes training labels that directly overlap the test period, while embargoing removes observations immediately after the test fold to reduce leakage caused by serial dependence. In contrast, walk-forward validation ensures that each test period occurs strictly after its training period, but it does not by itself eliminate overlapping-label leakage at the train/test boundary, so purging may still be required.

b) The Brier score is

$$BS = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2$$

with decomposition

$$BS = REL - RES + UNC$$

The overall observed default rate is

$$\bar{o} = \frac{5}{10} = 0.5$$

For reliability:

$$REL = \frac{1}{10}\left[5(0,4-0)^2 + 5(0,6-1)^2\right] = 0,16$$

$$\downarrow$$

$$REL = \frac{1}{N}\sum_k n_k(\bar{p}_k - \bar{o}_k)^2$$

For resolution:

$$RES = \frac{1}{N}\sum_k n_k(\bar{o}_k - \bar{o})^2 = \frac{1}{10}\left[5(0-0,5)^2 + 5(1-0,5)^2\right] = 0,25$$

For uncertainty:

$$UNC = \bar{o}(1-\bar{o}) = 0,5(1-0,5) = 0,25$$

Hence:

$$\boxed{BS = REL - RES + UNC = 0,16 - 0,25 + 0,25 = 0,16}$$

The model has perfect discrimination because every applicant who defaults receives a probability of 0.6, while every applicant who does not default receives a probability of 0.4. Therefore, all defaulters are ranked above all non-defaulters, corresponding to AUC = 1.

However, the model is poorly calibrated. The applicants assigned a 40% default probability actually default 0% of the time, so their risk is overstated, potentially causing excessive pricing or unnecessarily low credit limits. The applicants assigned a 60% probability actually default 100% of the time so their risk is understated, potentially leading to insufficient pricing and excessive credit exposure. Thus, perfect ranking does not imply that the predicted probabilities are economically correct.

A4| Using a first order Taylor expansion around the current parameter vector θ_t:

$$L(\theta_t + \Delta) \approx L(\theta_t) + \nabla L(\theta_t)^T \cdot \Delta$$

The gradient $\nabla L(\theta_t)$ points in the direction of steepest increase of the loss. Therefore, to decrease the loss locally, we choose Δ in the opposite direction:

$$\Delta = -\alpha \nabla L(\theta_t),$$

where α is the learning rate.

Since $\left[\theta_{t+1} = \theta_t + \Delta = \theta_t - \alpha \nabla L(\theta_t)\right]$

This is the gradient descent update rule.

i) If α is too large, gradient descent can overshoot the minimum, causing oscillation or even divergence. The descent lemma guarantees a decrease in the loss when $0 < \alpha \leq \frac{1}{L}$. Therefore, when $\alpha > \frac{1}{L}$, this guarantee no longer holds and a step may increase rather than decrease the loss.

ii) If α is too small, each update moves only a very short distance in the descent direction. The algorithm may still decrease the loss, but convergence becomes unnecessarily slow and requires many iterations.

iii) Feature scaling generally decreases the condition number of the loss surface by making the curvature in different parameter directions more similar. The loss surface therefore becomes less elongated and better conditioned. ~~(This reduces oscillation)~~ Thus, a learning rate α that is large enough to make progress in flatter directions is less likely to overshoot in steeper directions, so a wider range of α values can be used safely.

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